
Adaptive Data Dropout: Towards Self-Regulated Learning in Deep Neural Networks

Amar Gahir, Varshil Patel, Shreyank N Gowda
School of Computer Science, University of Nottingham, Nottingham, NG8 1BB, UK
`shreyank.narayanagowda@nottingham.ac.uk`

Abstract: Deep neural networks are typically trained by uniformly sampling large datasets across epochs, despite evidence that not all samples contribute equally throughout learning. Recent work shows that progressively reducing the amount of training data can improve efficiency and generalization, but existing methods rely on fixed schedules that do not adapt during training. In this work, we propose Adaptive Data Dropout, a simple framework that dynamically adjusts the subset of training data based on performance feedback. Inspired by self-regulated learning, our approach treats data selection as an adaptive process, increasing or decreasing data exposure in response to changes in training accuracy. We introduce a lightweight stochastic update mechanism that modulates the dropout schedule online, allowing the model to balance exploration and consolidation over time. Experiments on standard image classification benchmarks show that our method reduces effective training steps while maintaining competitive accuracy compared to static data dropout strategies. These results highlight adaptive data selection as a promising direction for efficient and robust training. Code: <https://github.com/userasg/Adaptive-Data-Dropout>

Keywords: Efficient AI, Data Dropout, Image Classification

1. Introduction

Advances in deep learning have consistently come at the cost of substantial computational resources. Each year, the computational cost of the most notable models multiplies by over 4 times [1]. This trend raises significant challenges for real-world deployment, scalability, and environmental sustainability [2]. While a large body of work has focused on improving model efficiency through architectural design, such as MobileNet [3,4], EfficientNet [5], and EfficientFormer [6], as well as techniques like pruning [7], quantization [8], and knowledge distillation [9], relatively less attention has been paid to improving the efficiency of the training process itself.

Standard training pipelines treat all samples equally across all epochs, implicitly assuming that every data point contributes uniformly throughout learning. However, prior work has shown that this assumption is flawed, as some samples provide diminishing returns once they are learned, while others remain informative for longer [10]. Inspired by this observation, recent work introduced Progressive Data Dropout (PDD) [11], a simple yet effective approach that progressively reduces the amount of training data across epochs. By focusing on informative samples early and revisiting the full dataset later, PDD achieves substantial reductions in effective training cost while maintaining or improving generalization.

Despite its effectiveness, PDD relies on pre-defined or pre-computed data schedules that remain fixed throughout training. These schedules, whether difficulty-based or stochastic, do not adapt to the evolving state of the model. As a result, they may either drop data too aggressively, leading to underfitting, or retain redundant samples longer than necessary, reducing potential efficiency gains. More fundamentally, fixed schedules fail to capture a key aspect of human learning: the ability to dynamically adjust study strategies based on ongoing performance.

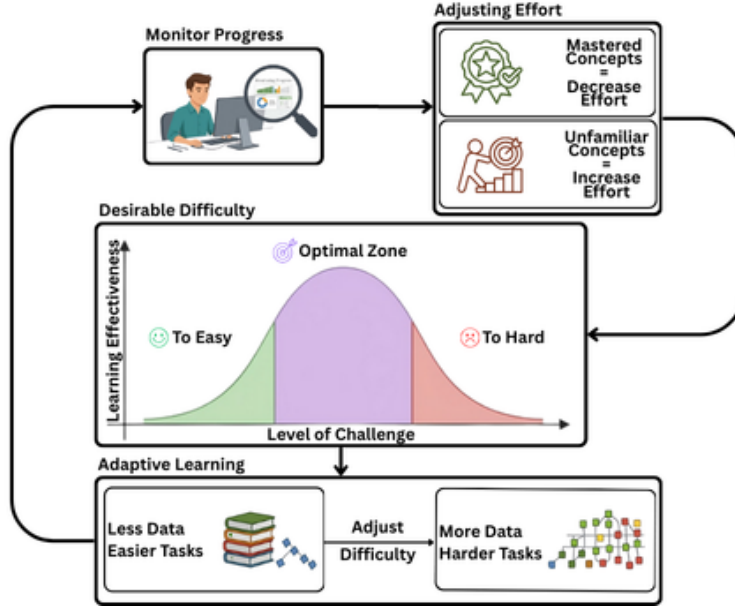


Figure 1. Adaptive learning balances challenge by dynamically adjusting effort based on performance feedback.

Human learning is inherently adaptive. Learners monitor their progress and regulate effort accordingly, allocating more attention when performance deteriorates and reducing effort when concepts are well understood. This process, often described as self-regulated learning [12], is closely linked to the notion of desirable difficulties [13], where learning is most effective when tasks are neither too easy nor too difficult. The optimal level of challenge is not static but evolves as the learner improves, requiring continuous adjustment of learning conditions [14]. These insights suggest that efficient learning systems should not rely on fixed curricula, but instead adapt the level of difficulty and data exposure over time. Figure 1 illustrates this visually.

In this work, we propose *Adaptive Data Dropout*, a simple and general framework that extends PDD by introducing feedback-driven data scheduling. Rather than following a fixed dropout schedule, our method dynamically adjusts the subset of training data at each epoch based on observed learning progress. Concretely, we use changes in training performance as a signal to modulate the degree of data dropout, allowing the model to automatically increase or decrease data exposure as needed. This results in a training process that maintains an appropriate level of difficulty throughout learning, balancing efficiency and generalization without requiring pre-defined schedules.

Our approach retains the simplicity and compatibility of PDD, requiring no changes to model architectures or optimization procedures, while introducing adaptivity into the training loop. We demonstrate that adaptive data dropout consistently reduces effective training cost while maintaining competitive performance across architectures. These results highlight the importance of feedback-driven data selection and suggest a promising direction toward more efficient and human-like learning dynamics in deep neural networks.

2. Related Work

Improving the efficiency and generalization of deep neural networks has long been a central goal in machine learning. Existing work spans model-centric optimization, data-centric selection, and adaptive training strategies, all aiming to reduce redundant computation while preserving performance. Our work is most closely related to data-centric approaches, but differs by treating data usage as a feedback-driven, dynamically regulated process rather than a predefined or static one.

Model-Centric Efficiency and Training Cost. Much prior work improves efficiency through model design and compression, especially for inference. Pruning [7], quantization [8], and knowledge distillation [9] reduce model size and computation, while architectures such as MobileNet [3], EfficientNet [5], and EfficientFormer [6] optimize parameter efficiency and FLOPs. However, these methods typically assume a fixed

training process with uniform data usage across epochs. In contrast, our work focuses on *training efficiency*, asking whether the amount of data used during optimization can be dynamically regulated.

Data Selection, Pruning, and Subset Training. A complementary line of work studies whether all training samples are equally necessary. Data Diet [15], dataset pruning [16], influence-based methods [17,18], and coreset selection [19,20] show that compact subsets can often preserve full-dataset performance. Yet these methods usually require pre-processing, auxiliary models, repeated scoring, or sample-level importance estimates, and often yield static subsets. Our approach instead adjusts data usage online during training, without constructing a fixed subset or explicitly scoring individual samples.

Curriculum Learning and Adaptive Sampling. Curriculum learning [21], hard example mining [22], and related adaptive sampling methods prioritize examples based on difficulty, loss, or uncertainty [10]. While effective, they often depend on predefined curricula or heuristics that may not transfer across datasets and architectures. They also focus mainly on *which* samples to prioritize. By contrast, our method operates at a coarser level, regulating *how much* data should be used at each stage based on training dynamics.

Data Dropout and Progressive Training Strategies. Recent data dropout methods reduce training cost by dropping samples during optimization. Data dropout [23], Dynamic Training Data Dropout [24], and Progressive Data Dropout (PDD) [11] demonstrate that effective training cost can be reduced without substantial accuracy loss. However, these approaches typically rely on fixed heuristics or predefined schedules, which may remove data too aggressively or retain redundant samples for too long. Our work builds on this insight by introducing a feedback-driven mechanism that adapts data usage in response to training performance, rather than following a predetermined schedule.

Adaptive Learning and Feedback-Driven Training. The broader idea of adapting learning based on feedback is well established in cognitive science and education. Self-regulated learning [12], desirable difficulties [13], and the challenge point framework [14] emphasize that learning is most effective when difficulty is dynamically matched to learner progress. While these ideas have inspired machine learning research, they have rarely been operationalized as dynamic control over training data volume. Our approach bridges this gap by using training feedback to regulate data exposure, maintaining an appropriate level of challenge throughout learning.

3. Method

We introduce *Adaptive Data Dropout*, a feedback-driven extension of Progressive Data Dropout (PDD) [11]. Unlike PDD, which follows fixed or precomputed schedules, our method dynamically adjusts the amount of training data used at each epoch based on performance feedback. This allows the model to regulate data exposure over time, reducing redundant computation while maintaining learning difficulty.

3.1. Overview

Let $\mathcal{D}_0 = \{(x_i, y_i)\}_{i=1}^N$ denote the full training dataset. At each epoch t , we construct a subset $\mathcal{D}_t \subseteq \mathcal{D}_0$ for backpropagation. Following PDD, the final epoch uses the full training set, $\mathcal{D}_T = \mathcal{D}_0$.

Rather than determining $|\mathcal{D}_t|$ from a fixed schedule, Adaptive Data Dropout updates the subset size using a feedback signal:

$$\Delta_t = \mathcal{A}_t - \mathcal{A}_{t-1},$$

where $\mathcal{A}_t$ is the training accuracy at epoch t . Positive feedback indicates learning progress and permits stronger data reduction, while stagnant or negative feedback increases data exposure. An overview is shown in Figure 2.

3.2. Variant 1: Adaptive Decay (Adaptive- α)

The first variant adapts the decay parameter α_t in a scalar-based dropout schedule:

$$|\mathcal{D}_t| = f(t; \alpha_t) \cdot N,$$

where f is a monotonically decreasing function, such as exponential, logarithmic, or inverse linear, and α_t controls the decay rate.

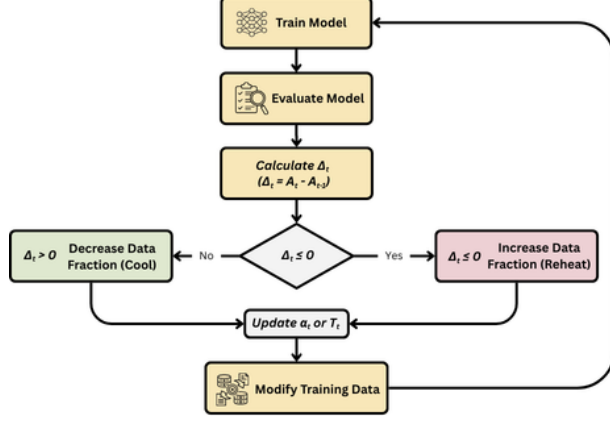


Figure 2. Adaptive Data Dropout adjusts training data usage based on changes in model performance, increasing or decreasing data exposure to maintain learning difficulty.

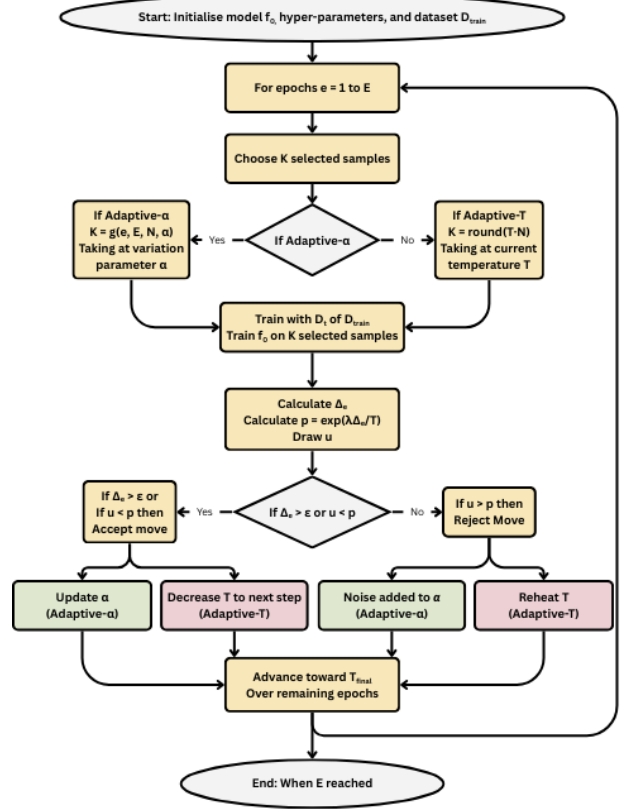


Figure 3. The two variants update either the decay parameter α or the keep-fraction T through an accept–reject mechanism that balances efficiency and learning stability.

At each epoch, α_t is updated through a stochastic acceptance rule. If $\Delta_t > 0$, a more aggressive decay is accepted by increasing α_t , thereby reducing the dataset size. If $\Delta_t \leq 0$, the update is accepted only probabilistically; otherwise, α_t is decreased to increase data exposure in later epochs. Thus, the model reduces redundant data when learning progresses, but revisits more data when performance stagnates.

3.3. Variant 2: Adaptive Keep-Fraction (Adaptive- T)

The second variant directly controls the retained data fraction $T_t \in (0, 1]$:

$$|\mathcal{D}_t| = T_t N.$$

We define a base trajectory $T_{\text{base}}(t)$, for example using a logarithmic schedule, but allow the actual trajectory to deviate according to feedback. If $\Delta_t > \delta$, where δ is a small threshold, the method follows the base schedule:

$$T_{t+1} = T_{\text{base}}(t+1).$$

Otherwise, a stochastic acceptance rule is applied. If the update is rejected, the system is *reheated* by increasing the keep-fraction:

$$T_{t+1} = \min(T_0, \gamma \cdot T_t),$$

where $\gamma > 1$ is a reheating factor. This allows recovery from overly aggressive data reduction by temporarily increasing data exposure.

3.4. Subset Sampling and Effective Epochs

At each epoch, $\mathcal{D}_t$ is formed by uniformly sampling $|\mathcal{D}_t|$ examples from $\mathcal{D}_0$ without replacement. Sampling is repeated independently across epochs to preserve diversity while reducing computational cost.

We measure training efficiency using *Effective Epochs* (EE) [11]. Since backpropagation is applied only to the selected subset, the total number of effective epochs is:

$$\text{Effective Epochs} = \frac{\sum_{t=1}^T |\mathcal{D}_t|}{N}.$$

This gives the equivalent number of full-dataset passes and provides a hardware-independent measure of training cost.

3.5. Discussion and Theoretical Insight

Adaptive Data Dropout generalizes PDD by replacing fixed schedules with feedback-driven updates. Conceptually, it resembles self-regulated learning: when performance improves, the model can tolerate less data and train more efficiently; when performance stagnates, it increases exposure to stabilize learning.

From an optimization perspective, changing $|\mathcal{D}_t|$ modulates stochastic gradient noise. Using less data increases gradient variance and can encourage exploration, while using more data reduces variance and supports stable convergence. The stochastic acceptance and reheating mechanisms help avoid overly aggressive data reduction, enabling the method to alternate between efficient exploration and consolidation. Although a full theoretical characterization remains open, this view suggests that adaptive data exposure acts as an implicit regularizer, shaping both optimization and generalization.

4. Experimental Analysis

4.1. Datasets

We evaluate our method on three standard image classification benchmarks, namely CIFAR-10 [25], CIFAR-100 [25] and ImageNet [26]. The CIFAR-10 and CIFAR-100 datasets each contain 60,000 color images of size 32×32 , with 50,000 training images and 10,000 test images. CIFAR-10 consists of 10 object categories, while CIFAR-100 contains 100 finer-grained classes. These datasets are widely used for evaluating compact and mid-scale architectures. We also evaluate on ImageNet (ILSVRC2012), which contains over 1.28 million training images and 50,000 validation images spanning 1,000 categories. Compared to CIFAR, ImageNet provides substantially greater scale and diversity, making it a strong benchmark for assessing both generalization and training efficiency.

4.2. Implementation Details

Adaptive Data Dropout requires only minimal changes to standard training pipelines: we keep the model architecture, loss function, optimizer, batch normalization, learning-rate schedule, and augmentation policy unchanged, modifying only the epoch-wise selection of training subsets. For all data-reduction variants, the data loader is refreshed each epoch using the current sampled subset. We evaluate on common image classification architectures, including MobileNetV2 [4], EfficientNet [5], EfficientFormer [6], ResNet [27], and ViT-B [28]. ImageNet experiments follow official implementation details and prior optimization settings, while CIFAR experiments use AdamW, StepLR, and standard augmentations. We test two architecture-agnostic variants: **Adaptive- α** , which adjusts the subset decay parameter, and **Adaptive- T** , which directly controls the keep-fraction. In both cases, the final epoch uses the full dataset as a revision phase [11].

4.3. Supervised Image Classification

Table 1 compares EfficientNet-B0 [5], MobileNet-V2 [4], ResNet-50 [27], EfficientFormer-L1 [6], and ViT-B-MAE [29] across CIFAR-10, CIFAR-100, ImageNet, and ImageNet-pretrained transfer settings. We report baseline training, DBPD [11] as a strong static data dropout baseline, and our two variants, Adaptive- α and Adaptive- T , to assess whether feedback-driven scheduling improves the accuracy-cost trade-off over fixed schedules. Across datasets and architectures, static dropout already reduces training cost, but adaptive scheduling further improves robustness and efficiency by adjusting data exposure according to learning progress. Figure 4 illustrates this trade-off: Adaptive Data Dropout achieves competitive or improved

Table 1. Accuracy (%) and effective epochs reported as *accuracy/effective epochs*. All methods besides ViT-MAE are trained from scratch unless otherwise mentioned. ViT-MAE uses the publicly available checkpoint [29]. All results have been conducted following official implementations on timm [30]. Best results in **blue** and second best in **green**. The last column shows the gains in accuracy and drop in effective epochs of Adaptive Data Dropout relative to the baseline.

	Baseline	DBPD	Adaptive- α	Adaptive- T	Acc. gain (%)	EE saved
<i>CIFAR10</i>						
EfficientNet-B0	88.21 / 30	87.82 / 5.86	85.69 / 7.45	85.28 / 6.36	-2.52	4.72 ×
MobileNet-V2	85.49 / 30	85.39 / 6.42	83.08 / 8.76	80.79 / 6.14	-2.41	4.89 ×
ResNet-50	84.85 / 30	85.61 / 7.08	83.82 / 8.48	82.64 / 6.24	-2.21	4.81 ×
EfficientFormer-L1	80.32 / 30	81.17 / 5.82	80.75 / 7.21	80.67 / 6.90	0.43	5.15 ×
<i>CIFAR100</i>						
EfficientNet-B0	66.32 / 200	67.15 / 24.8	66.51 / 23.94	63.67 / 21.13	0.19	8.36 ×
MobileNet-V2	61.25 / 200	62.85 / 29.6	63.14 / 25.61	62.99 / 22.62	1.89	7.84 ×
ResNet-50	59.81 / 200	60.13 / 16.26	59.67 / 16.11	59.41 / 13.78	-0.14	14.51 ×
EfficientFormer-L1	55.91 / 200	57.62 / 30	58.11 / 31.31	55.95 / 24.67	2.20	8.11 ×
<i>CIFAR100 finetuned from ImageNet</i>						
EfficientNet-B0	83.31 / 200	83.95 / 25.4	83.98 / 26.1	84.11 / 20.8	0.80	9.62 ×
MobileNet-V2	74.10 / 200	74.35 / 22.4	74.61 / 25.3	74.39 / 19.5	0.51	10.26 ×
ResNet-50	78.59 / 200	79.92 / 21	80.17 / 23.4	79.97 / 17.8	1.58	11.24 ×
EfficientFormer-L1	85.30 / 200	86.74 / 36.6	87.28 / 39.8	86.85 / 33.2	1.98	6.02 ×
ViT-B-MAE	86.81 / 200	85.92 / 52.2	86.94 / 55.8	85.55 / 43.5	0.13	4.60 ×
<i>ImageNet</i>						
EfficientNet-B0	77.10 / 350	77.45 / 111.2	78.15 / 134.2	77.71 / 105.5	1.05	3.32 ×
MobileNet-V2	71.60 / 250	68.42 / 54.5	70.82 / 53.9	67.59 / 45.8	-0.78	5.46 ×
ResNet-50	75.04 / 100	76.21 / 33.0	76.82 / 35.5	76.25 / 31.5	1.78	3.17 ×
EfficientFormer-L1	79.11 / 300	79.24 / 91.5	79.45 / 95.8	78.87 / 85.7	0.34	3.50 ×
ViT-B-MAE	83.10 / 100	83.31 / 35.4	83.81 / 37.8	83.06 / 27.5	0.71	3.64 ×

accuracy with fewer effective epochs, placing the adaptive variants in more favorable regions of the Pareto frontier than standard training and fixed dropout schedules.

4.4. Comparison to State-of-the-Art

We compare Adaptive Data Dropout against strong baselines for efficient training and data selection, including DataDiet (ELN and Forget) [31], IES [32], Early Stopping, and InfoBatch [33]. We follow the official implementations for all baselines and use effective epochs to evaluate efficiency. This comparison is designed to assess whether adaptive control of data exposure provides benefits beyond fixed subset schedules, heuristic importance estimation, or early termination strategies. The results are reported on CIFAR-100 fine-tuning from ImageNet under a common training setup.

Table 2. Comparison with strong active learning baselines on CIFAR-100. Best results are shown in **blue** and second best in **green**. DDE corresponds to DataDiet ELN and DDF corresponds to DataDiet Forget.

Model	Baseline	DDE	DDF	IES	Early Stopping	InfoBatch	DBPD	Adaptive- α	Adaptive- T
ResNet-50	78.5 / 200	78.6 / 141	79.1 / 142	79.5 / 157	74.3 / 121	80.4 / 145	80.0 / 21	80.2 / 23.4	80.0 / 17.8
EfficientNet	83.3 / 200	82.7 / 139	83.0 / 139	83.7 / 153	76.6 / 132	84.0 / 141	84.0 / 25.1	84.0 / 26.1	84.1 / 20.8
EfficientFormer-L1	85.3 / 200	84.8 / 142	85.0 / 140	85.8 / 155	78.2 / 125	86.6 / 143	86.8 / 36.6	87.3 / 39.8	86.9 / 33.2

4.5. Ablation Study

To better understand the behavior of Adaptive Data Dropout, we conduct an ablation study along several dimensions, including matched-step baseline comparisons, sensitivity to adaptive control parameters, shortened training schedules, and the effect of revision. Unless otherwise stated, all ablations are performed using EfficientNet-B0 trained from scratch on CIFAR-100.

Are the gains simply due to fewer effective epochs? To ensure that any improvement is not merely a consequence of reduced optimization steps, we train a baseline EfficientNet-B0 for the same number of

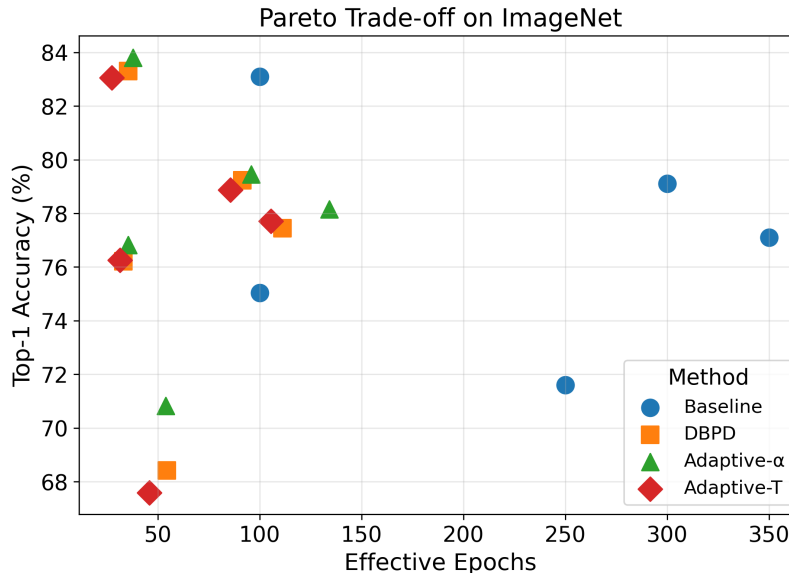


Figure 4. Pareto trade-off between Top-1 accuracy and effective epochs on ImageNet. Each point represents a model, with colors indicating architectures and markers denoting methods. Adaptive Data Dropout consistently achieves a better accuracy–efficiency trade-off, achieving competitive accuracy while using substantially fewer effective training epochs.

Table 3. Baseline trained to match the effective number of epochs (EE) used by dropout variants.

Variant	EE	Baseline (%)	Accuracy (%)
Adaptive- α	23.9	63.18	66.51
Adaptive- T	21.1	61.82	63.67

effective epochs as each dropout variant. This comparison isolates whether performance gains arise from the schedule itself rather than from training less. We report these results in Table 3.

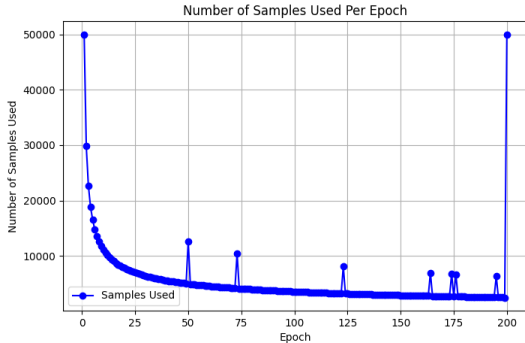
Can we shorten training and still retain performance? To assess performance under constrained training budgets, we repeat all dropout variants under shorter training schedules and compare them to baselines trained for the same number of epochs. This experiment evaluates whether adaptive data exposure remains useful when optimization time is limited.

4.6. Dynamics of Adaptive Reheating

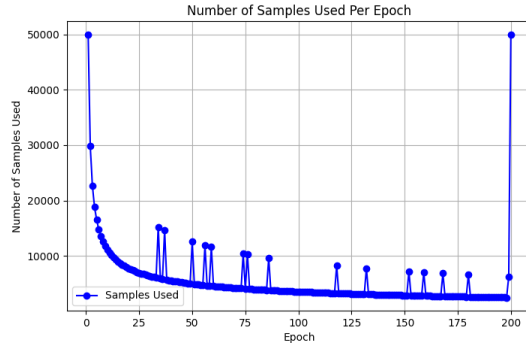
To better understand the behavior of Adaptive Data Dropout, we analyze how the number of training samples evolves over epochs. In particular, we examine whether the proposed feedback-driven mechanism exhibits the intended “reheating” behavior, where data exposure is increased when learning stagnates or deteriorates. Figure 5 visualizes the number of samples used per epoch for two representative models. We observe a consistent pattern across both variants: the training process initially reduces the dataset size as performance improves, reflecting increasing confidence and redundancy in the data. However, at several points during training, sharp increases in the number of samples occur. These spikes correspond to periods where the improvement in training accuracy slows down or reverses, triggering the adaptive mechanism to increase data exposure. This behavior closely mirrors the notion of self-regulated learning, where learners allocate more effort when performance deteriorates and reduce effort when learning is progressing well. Importantly, these reheating events are not pre-scheduled but emerge naturally from the feedback signal, demonstrating that the model dynamically adjusts its learning difficulty over time. Overall, these results provide empirical evidence that Adaptive Data Dropout does not follow a fixed curriculum, but instead continuously adapts data usage in response to training dynamics. This ability to recover from overly aggressive data reduction is a key advantage over static schedules such as Progressive Data Dropout.

Table 4. Accuracy (%) and effective epochs reported as *accuracy/effective epochs* for CIFAR-100 with EfficientNet-B0, trained for 25 and 50 epochs from scratch.

Model	Baseline	DBPD 0.7	Adaptive- α	Adaptive- T
(50 epochs)				
EfficientNet-B0 (C100)	63.65 / 50	67.62 / 18.64	67.95 / 20.71	66.58 / 14.38
(25 epochs)				
EfficientNet-B0 (C100)	63.32 / 25	66.94 / 14.02	66.85 / 15.24	66.07 / 10.51



(a) EfficientNet-B0



(b) ResNet-50

Figure 5. Number of training samples used per epoch under Adaptive Data Dropout. The model progressively reduces data usage but intermittently increases it in response to stagnation or drops in performance, demonstrating adaptive reheating behavior.

5. Limitations and Future Work

Adaptive Data Dropout improves over fixed data schedules, but it has several limitations. First, it introduces control hyperparameters, such as update thresholds, decay functions, and reheating factors, which may require tuning across datasets, architectures, and optimizers. Second, because adaptation relies on training feedback, noisy or unstable signals, especially early in training, may lead to suboptimal updates. More robust signals, such as smoothed accuracy, loss trends, or uncertainty estimates, could improve stability.

Third, our evaluation focuses on standard image classification benchmarks. Further validation is needed for detection, segmentation, video understanding, and other settings with different training dynamics. Finally, although the method is more flexible than fixed schedules, it still requires choosing an initial schedule family or operating range. Future work could develop more autonomous controllers, including reinforcement learning or meta-learning policies, to learn data scheduling directly from training signals. Such methods could also adapt other aspects of training, such as augmentation, sampling, or learning-rate schedules.

6. Conclusion

We introduced *Adaptive Data Dropout*, a feedback-driven extension of Progressive Data Dropout that replaces fixed data schedules with adaptive data exposure. Inspired by self-regulated learning, our method adjusts the amount of training data used during optimization according to observed learning progress. We proposed two variants, Adaptive- α and Adaptive- T , which respectively control the decay parameter and the keep-fraction of the training data. Across image classification benchmarks and architectures, Adaptive Data Dropout improves the accuracy-efficiency trade-off by reducing effective training cost while maintaining competitive performance. More broadly, our results suggest that training efficiency can be improved not only through better architectures or optimizers, but also by making data usage responsive to the model’s learning dynamics. This points toward a promising direction for self-regulating training systems that adapt their exposure to data over time.

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